

Mudasser Afzal

Data Scientist | Machine Learning Engineer

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PROFESSIONAL SUMMARY

I'm a Data Scientist and ML Engineer with 5+ years of experience building systems that turn messy, unstructured data into something that actually works in production. My background spans NLP pipelines, transformer-based classifiers, and recommendation systems across startups and scale-ups in Pakistan, remotely, and now in Berlin. At Intapp, I work on a knowledge graph for reranking of results, an agentic system to extract financial information from text and document parsing inside a RAG system, which is exactly the kind of problem I find most interesting: making legal and financial text genuinely usable for downstream applications. More recently, I've been going deep on agentic AI, building multi-agent LangGraph systems, a self-correcting coding agent with human-in-the-loop review, and a live hallucination detection.

TECHNICAL SKILLS

Machine learning: Classification, Regression | Clustering | Ensemble Methods | Recommendation Systems

Deep learning: Neural Networks (CNN | RNN | LSTM) | Transformers (BERT | LUKE) | Transfer Learning | Large Language Models (LLMs) | Fine-tuning (LoRA | PEFT)

NLP & text mining: Named Entity Recognition (NER) | Relation Extraction | Text Classification | Sentiment Analysis | Embeddings | Prompt Engineering | RAG | Layout Extraction | Structured Outputs (Pydantic)

Agentic AI & GenAI: OpenAI / Azure OpenAI API | HuggingFace | LangChain | LangGraph | LangSmith | LLM Observability | Multi-agent Systems | Human-in-the-Loop (HITL) Patterns | ChromaDB | Pinecone

Programming & frameworks: Python (PyTorch | TensorFlow | Keras | scikit-learn | NLTK | spaCy | NumPy | Pandas) | MySQL | Flask | Django | FastAPI

Data analysis & visualisation: Pandas | Seaborn | Matplotlib | Plotly | Feature Engineering | Data Cleaning

Cloud & MLOps: AWS (EC2 | S3 | Lambda | SageMaker) | Azure (ML Studio | Document Intelligence) | Docker | MLflow | Git | Bitbucket | Railway

Development & collaboration: Jupyter | Google Colab | Jira | Confluence | Agile | Cursor | Claude

PROFESSIONAL EXPERIENCE

Machine Learning Engineer (Working Student)

12/2023 – Present

Intapp Inc, Berlin, Germany

Product: Business Analytics through **firmographic data**.

- Designed and implemented a Named Entity Recognition (NER) system using the Luke model, extracting 5 key business-related entities, achieving 88% accuracy on the test set.
- Led the development of a **relation extraction model** for identifying strategic business relationships (e.g., partnerships, acquisitions), improving model accuracy by 20%.
- Currently working on a **Document Parsing** project as part of a **Retriever-Augmented Generation (RAG)** model, improving document parsing accuracy and **data retrieval** efficiency.

Technologies: Python, Transformers, Luke, NER, Pandas, ChatGPT, NLP, Azure, Data Mining, RPC

Data Scientist

03/2023 – 10/2023

3S Solutions, Lahore, Pakistan

Product: Sibme - AI-Powered Teacher Evaluation Tool.

- Implemented **ChatGPT**-powered mentoring systems that improved user engagement, increasing interaction time by 40%.
- Collaborated on AI-driven **performance tracking** tools that identified key teacher metrics, leading to a 30% improvement in client satisfaction and retention.

- Delivered actionable insights for performance improvements, which contributed to a **20% boost** in positive feedback from educational institutions.

Technologies: Python, EC2, AWS, Transformers, OpenCV, YOLO, ChatGPT, Pandas, AWS Lambda

Machine Learning Scientist

05/2022 – 12/2022

Turing, Remote

Project: AI Vetting System for Shortlisting Candidates

- Enhanced the **automated resume parsing pipeline**, increasing clean data extraction from 2% to 13%, providing higher-quality data for AI models.
- Conducted **feature engineering** on candidate data, improving pairwise ranking scores by 2%, significantly optimising the candidate selection process.

Technologies used: Python | Feature Engineering | Pandas | Git | CentOS

Data Scientist

08/2021 – 08/2022

Algo.ai, Lahore, Pakistan

Project: Lost Sale Detection Project for Vodafone Clients.

- Developed a recommendation system that increased sales by **20%**, enabling stores to suggest alternative products to customers in real-time.
- Created a data analysis tool using **unsupervised learning techniques**, which allowed non-technical clients to perform custom data analysis and share key insights, improving **customer satisfaction by 35%**.

Technologies: Python, Pandas, Jupyter Notebook, NLP, Git, Django

Machine Learning Engineer

02/2021 – 07/2021

Senarios, Lahore, Pakistan

Project: Brand Recommendations & Customer Interaction Automation

- Built a Graph Neural Network (GNN) for brand recommendations, achieving a 15% increase in user engagement by providing personalised suggestions based on demographic data.
- Proposed an automated BERT-based classifier for call transcriptions that increased customer outreach efficiency by 25% and streamlined communication efforts.

Technologies: Python, Jupyter Notebook, Pandas, BERT, NLP, Git, Heroku, Flask API

EDUCATION

Brandenburg Technical University, Cottbus, Germany

10/2023 – 07/2026

Master of Science – Artificial Intelligence

Focus: Explainable AI, Information Retrieval, Data Mining

Thesis: Script-Aware CNN-FPN Transformer for Offline Handwritten Urdu OCR

Information Technology University, Lahore, Pakistan

10/2019 – 06/2023

Master of Science – Data Science

Focus: Deep Learning, Machine learning, Data Mining

Pakistan Institute of Engineering and Applied Science, Pakistan

11/2015 – 06/2019

Bachelor of Science – Computer and Information Science

Focus: Artificial Intelligence, Computational Intelligence, Pattern Recognition

Thesis: Image Segmentation using Autoencoders